

TAGBILARAN, Philippines

WMO: 986440

Lat: 9.664N

Lon: 123.853E

Elev: 38

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.9	73.3	68.5	104.7	75.3	69.7	109.2	76.4	14.7	80.1	12.8	81.6	2.1	0	0.330

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.0	92.9	81.3	91.9	81.1	91.0	80.9	82.9	89.7	82.4	89.4	81.9	88.7	5.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	161.7	87.4	80.5	158.5	87.3	80.0	155.8	86.8	46.9	89.7	46.3	89.7	45.7	89.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	10.8	9.3	DB	68.0	95.0	4.0	2.9	65.2	97.1	62.8	98.8	60.6	100.5	57.6	102.6	(4)
			WB	67.0	84.6	3.1	1.0	64.8	85.3	63.0	85.9	61.2	86.4	58.9	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	80.3	80.6	81.5	82.9	84.0	83.2	82.9	83.6	83.3	82.7	82.1	81.3
	DBStd	2.17	1.89	2.03	2.10	1.78	1.59	1.80	1.83	2.00	2.00	1.77	1.59	1.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11817	939	857	975	987	1055	997	1020	1041	998	1014	963	971
	CDD65	6342	474	437	510	537	590	547	555	576	548	549	513	506
	CDH74	69325	4213	4156	5294	6245	7300	6369	6256	6772	6355	6047	5353	4965
	CDH80	24918	1115	1279	1861	2519	3108	2427	2242	2658	2397	2119	1728	1465
Wind	WSAvg	3.5	3.9	3.7	3.7	3.2	3.3	3.2	3.7	4.0	3.8	3.3	3.1	3.5
Precipitation	PrecAvg	55.6	4.7	3.3	3.9	3.1	3.5	5.7	5.7	4.7	4.9	6.8	7.2	5.4
	PrecMax	78.1	16.6	9.6	33.0	9.2	8.3	11.9	15.0	9.5	12.9	15.2	17.6	14.3
	PrecMin	37.5	0.3	0.0	0.1	0.0	0.0	0.4	0.4	0.4	1.0	1.2	0.4	1.3
	PrecStd	11.1	3.7	2.4	5.2	2.4	2.4	2.9	2.8	2.2	2.8	2.8	3.5	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	91.5	92.5	93.3	93.6	93.5	92.5	93.9	93.4	92.5	91.2	90.6
		MCWB	79.1	79.5	79.9	81.1	81.5	81.8	81.6	81.6	81.3	81.7	81.0	80.6
	2%	DB	88.2	89.7	90.6	91.8	92.3	91.4	90.9	92.2	91.6	90.7	89.8	89.0
		MCWB	78.8	79.1	79.8	80.7	81.3	81.4	81.2	81.3	81.1	81.2	80.7	80.3
	5%	DB	87.0	88.2	89.2	90.6	91.0	89.9	89.5	90.7	90.2	89.3	88.6	87.7
		MCWB	78.6	78.6	79.6	80.4	81.1	81.1	80.8	80.8	80.7	80.9	80.5	79.9
	10%	DB	85.6	86.7	87.8	89.3	89.8	88.6	88.3	89.0	88.7	87.9	87.3	86.5
		MCWB	78.4	78.2	79.2	80.1	81.0	80.8	80.6	80.4	80.4	80.5	80.2	79.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	81.7	82.0	83.0	83.6	83.6	83.0	83.3	82.8	82.8	82.6	82.4
		MCDB	86.8	88.3	89.2	90.4	90.4	89.8	89.5	90.8	90.0	89.4	88.7	88.0
	2%	WB	80.2	80.5	81.2	81.9	82.6	82.5	82.0	82.2	81.9	81.9	81.7	81.4
		MCDB	85.9	87.2	88.3	89.4	89.8	89.1	88.7	89.6	88.8	88.6	87.8	87.0
	5%	WB	79.6	79.7	80.5	81.3	81.9	81.7	81.4	81.5	81.3	81.3	81.0	80.7
		MCDB	85.1	86.1	87.5	88.6	89.0	88.2	88.0	88.6	88.1	87.7	87.0	86.3
	10%	WB	78.8	78.9	79.7	80.6	81.3	81.0	80.8	80.8	80.7	80.7	80.5	80.0
		MCDB	84.2	85.1	86.4	87.8	88.3	87.3	87.2	87.6	87.4	86.9	86.3	85.3
Mean Daily Temperature Range	5% DB	MDBR	10.6	12.0	12.7	13.0	12.0	11.1	10.8	11.2	10.8	11.1	11.1	10.8
		MCDBR	12.8	14.6	14.8	14.8	13.5	13.0	12.5	13.5	13.1	13.0	12.9	12.6
		MCWBR	6.1	6.7	7.0	6.8	6.1	6.0	5.9	5.8	5.6	6.1	6.2	6.1
	5% WB	MCDBR	11.3	12.7	13.2	13.4	12.4	11.9	11.7	12.1	11.7	12.0	11.8	11.4
		MCWBR	6.3	7.0	7.1	7.0	6.4	6.3	6.2	6.1	5.8	6.2	6.4	6.4
Clear-Sky Solar Irradiance	taub	0.429	0.419	0.419	0.425	0.425	0.417	0.419	0.438	0.434	0.446	0.428	0.426	
	taud	2.443	2.459	2.465	2.445	2.461	2.493	2.488	2.417	2.423	2.388	2.452	2.455	
	Ebn at Noon	279	286	286	281	276	274	275	273	278	274	278	278	
	Edh at Noon	36	37	37	37	36	34	35	38	38	39	36	35	
All-Sky Solar Radiation	RadAvg	1310	1565	1822	1976	1818	1652	1574	1640	1598	1554	1472	1335	
	RadStd	162	159	153	89	117	92	93	113	139	103	123	120	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)		+0.27	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page